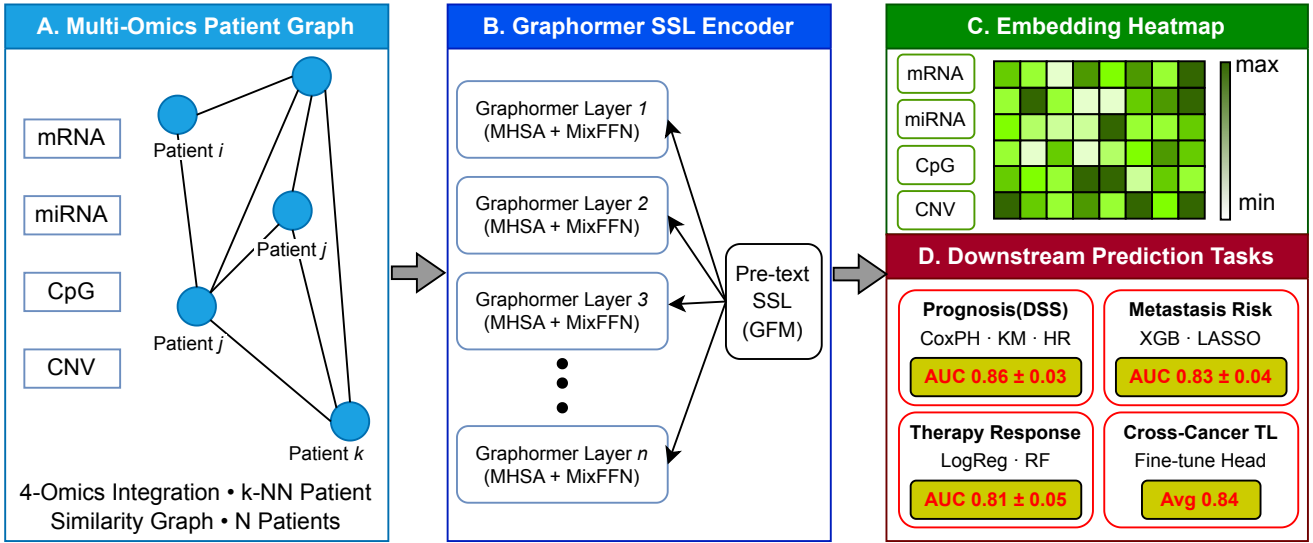

A. Multi-Omics Patient Graph
mRNA
miRNA
CpG
CNV
Patient i
Patient j
Patient j
Patient k
4-Omics Integration • k-NN Patient Similarity Graph • N Patients
B. Graphormer SSL Encoder
Graphormer Layer 1 (MHSA + MixFFN)
Graphormer Layer 2 (MHSA + MixFFN)
Graphormer Layer 3 (MHSA + MixFFN)
Graphormer Layer n (MHSA + MixFFN)
Pre-text SSL (GFM)
C. Embedding Heatmap
mRNA
miRNA
CpG
CNV
max
min
D. Downstream Prediction Tasks
Prognosis(DSS)
CoxPH · KM · HR
AUC 0.86 ± 0.03
Metastasis Risk
XGB · LASSO
AUC 0.83 ± 0.04
Therapy Response
LogReg · RF
AUC 0.81 ± 0.05
Cross-Cancer TL
Fine-tune Head
Avg 0.84